



Efficiency Enhancement in Fiber Laser Cutting Process: Compressed Air Usage and Parameter Optimization for 2mm Mild Steel

Group I2-08

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Dalgakıran Compressor founded in 1965 by Ömer Dalgakıran in a 25 m² workshop, now Turkey's largest industrial compressor manufacturer and exporter.

Operates in modern facilities with a total indoor area of over 70,000 m².

A dynamic team of over 900 employees carrying Turkish engineering to the global stage.

Exports to over 130 countries.

Network of 150+ dealers and widespread service points.

Ranked among Turkey's top 500 exporters.

Home to Turkey's first R&D center in the compressor sector, combining 60 years of experience with cutting-edge technology.

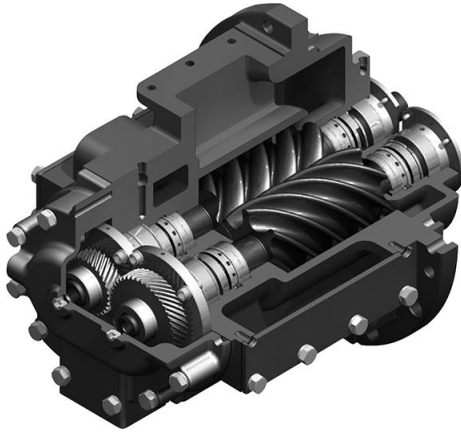
Production Facilities and Plant Layout

Office	Office	Final Test & Inspection Area		Warehouse						Receiving Department	
	Arc Line		Eagle Line		Warehouse	Canopy Production	Air Vessel Production	Air Oil Separation Vessel Production	Raw Frame Assembly	Prefabrication Assembly	
	Electrical Workshop	Sparrow Line		Owl Line						Dry Workshop	
Sponge Workshop		Paintshop							Warehouse		
	Office								Office		

Dalgakıran production facilities are organized according to lean manufacturing principles, from raw material entry to final product testing.

Product Range and Technical Competencies

Screw Compressors



Oil-Free Compressors

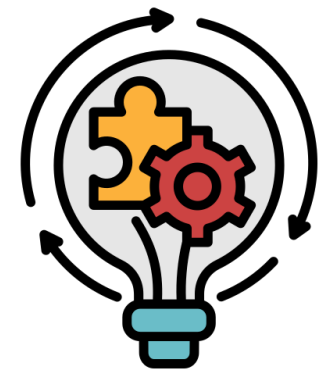
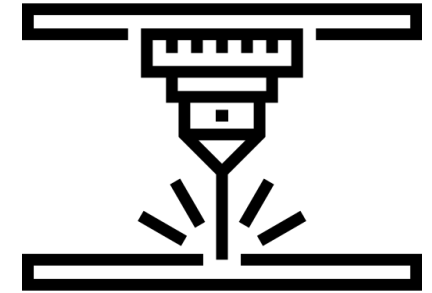


Reciprocating / Piston Compressors

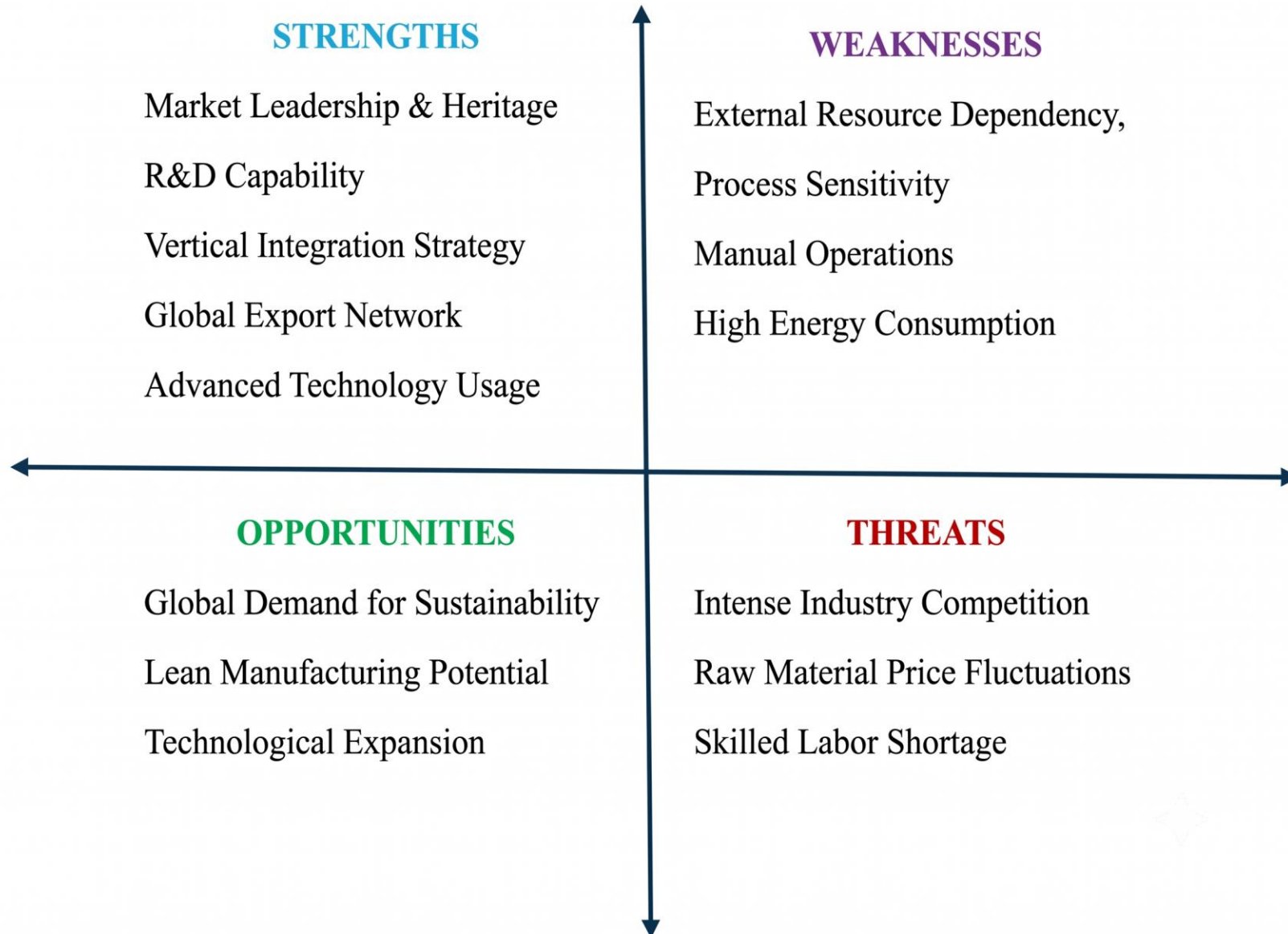


Production Culture and Technology

- Dalgakıran has a continuous improvement-oriented production culture.
- Key Technologies and Practices:
 - Kaizen culture
 - Daily GEMBA walks
 - SAP / ERP material tracking
 - SolidWorks design infrastructure
 - Semi-automatic welding systems
 - Fiber laser cutting technology



SWOT ANALYSIS



Project Selection Background and Matrix

Evaluation Criteria	Weight	1. Laser Cutting Air Conversion (Selected)	2. Robotic Welding Integration	3. Assembly Line Optimization
Investment Cost (Lower is better)	30%	9/10 (Only compressor investment)	2/10 (High robot cost)	10/10 (No investment required)
Return on Investment (ROI)	30%	10/10 (3.6 Months - Very Fast)	3/10 (2 Years - Slow)	6/10 (Uncertain/Long)
Ease of Implementation	20%	8/10 (Can be done without stopping machine)	4/10 (Line stoppage required)	5/10 (Physical relocation is hard)
Cost Saving Impact	20%	10/10 (Direct cash savings)	7/10 (Labor savings)	6/10 (Efficiency increase)
TOTAL SCORE	100%	9.3 (WINNER)	3.8	7.2

Selection Criteria:

- Investment cost
- Return on investment
- Ease of implementation
- Cost saving impact

Main Reason for Selection:

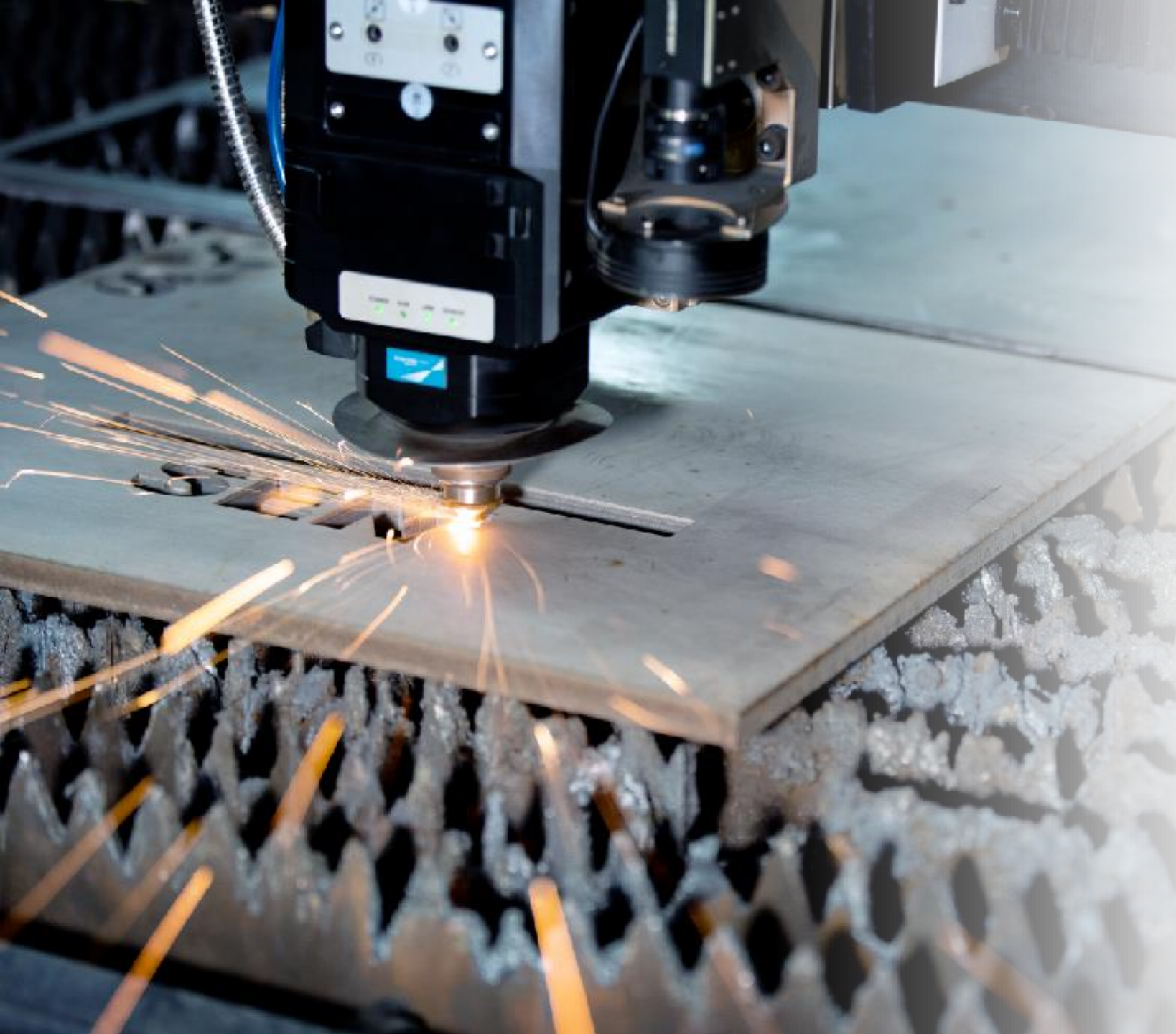
- Fast ROI
- Direct cost reduction
- Easy implementation
- High impact on production flow

Selected Project: Laser Cutting Improvement

The selected project focuses on improving the laser cutting process by:

- Replacing nitrogen with compressed air
- Optimizing cutting parameters
- Reducing dross formation
- Eliminating manual deburring
- Reducing unit cutting cost





Scope of the Project

- The scope of this project is to improve the laser cutting process by reducing assist gas cost, increasing cutting efficiency, and minimizing dross formation.
- The analysis focuses on the Bodor 12kW fiber laser machine and thin mild steel sheets used in compressor canopy production.

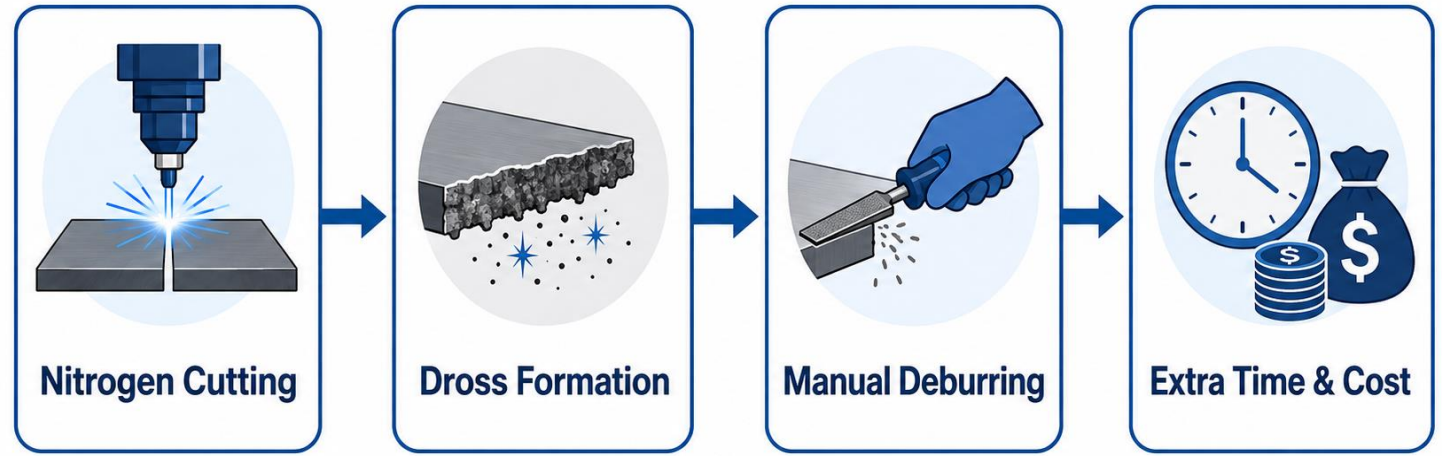
Main Variables:

- Assist gas type
- Gas pressure
- Focus position
- Cutting speed
- Nozzle type

Current Situation in Laser Cutting

In the current process:

- Nitrogen is used as the main assist gas
- Gas cost is high
- Dross formation occurs on the bottom surface
- Manual deburring is required
- Cutting parameters are not fully optimized



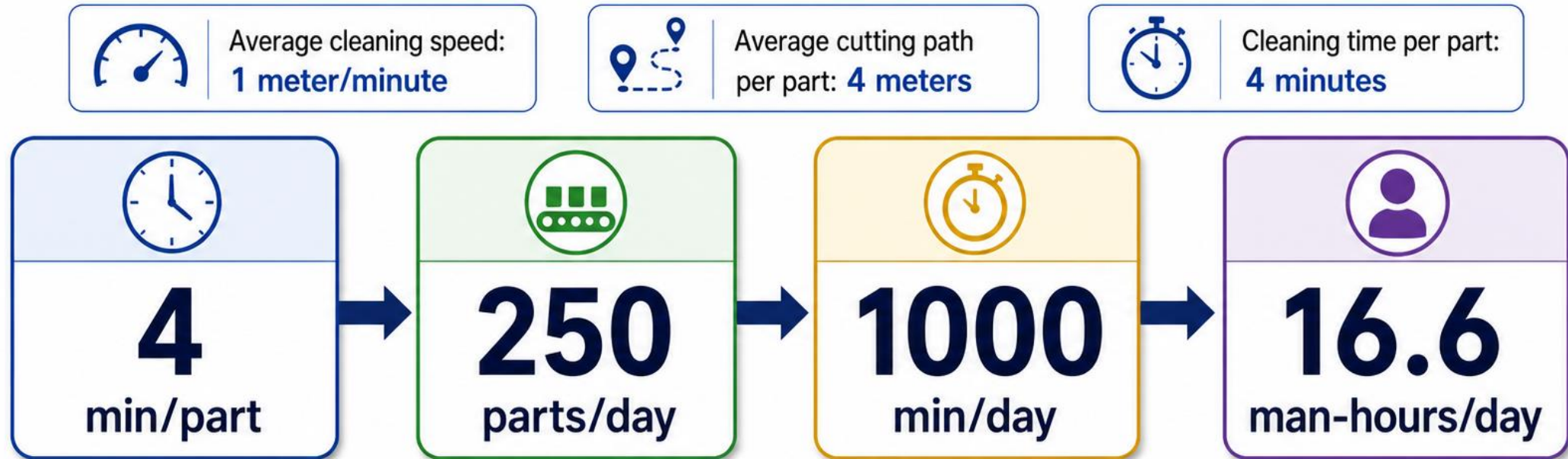
Main Problems Identified

In the current laser cutting process, three major problems were identified:

- **High Nitrogen Cost:** Externally supplied nitrogen creates a significant variable cost.
- **Dross Formation:** Incorrect focus position and insufficient gas pressure cause dross on the bottom surface.
- **Manual Deburring Requirement:** Dross formation creates extra labor time and non-value-added work.



Dross Cleaning Time Analysis



 $25 \text{ plates/day} \times 10 \text{ parts/plate} = 250 \text{ parts/day}$

 $250 \text{ parts} \times 4 \text{ min} = 1000 \text{ min/day} \approx 16.6 \text{ man-hours/day}$



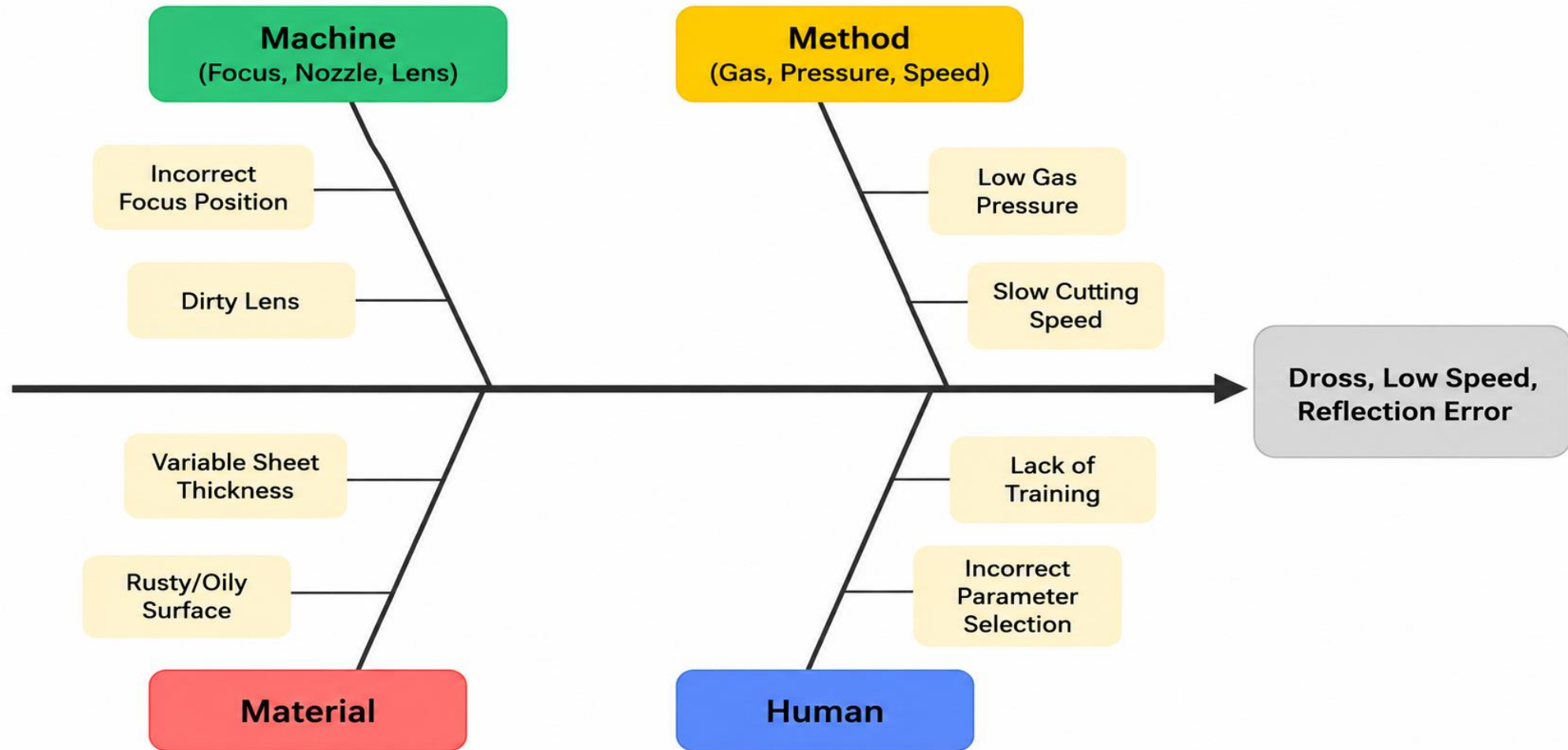
Main message: Almost two full-time operators are spent only for dross cleaning.

Pareto Analysis of Loss Sources

Problem / Loss Source	Annual Estimated Cost Impact (TL)	Share in Total (%)	Cumulative Percentage (%)
1. Assist Gas (Nitrogen) Cost	1,140,000 TL	62%	62%
2. Dross Cleaning Labor	500,000 TL	27%	89%
3. Downtime & Reflection Errors	120,000 TL	7%	96%
4. Cut Loss (Scrap Metal)	50,000 TL	3%	99%
5. Other (Consumables, etc.)	20,000 TL	1%	100%
TOTAL	1,830,000 TL	100%	

89% of total loss comes from **nitrogen cost** and **dross cleaning labor**.

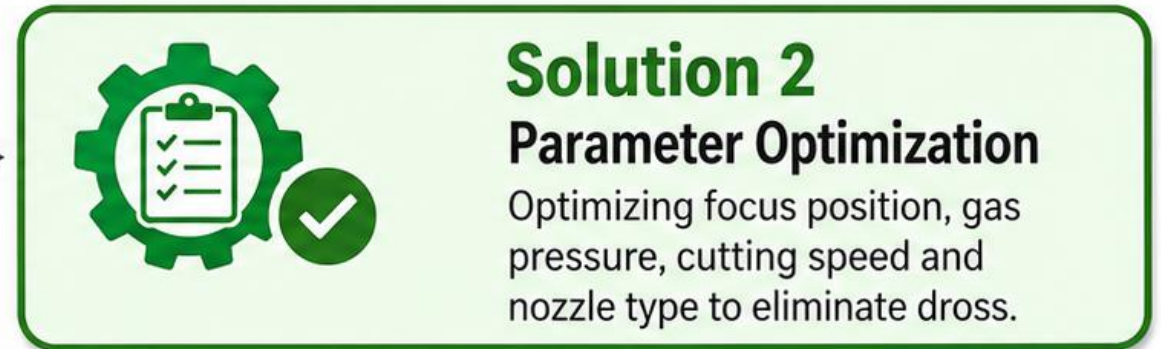
Root Cause Analysis (Fishbone Diagram)



- The root cause analysis shows that dross formation is mainly related to focus position, gas pressure and cutting speed. On the other hand, the high cost problem is directly related to nitrogen usage and external gas dependency.

Proposed Improvement Approach

To solve the identified problems, two improvement approaches were proposed:



Expected Impact:



Solution 1: Compressed Air Cutting

Compressed air is used as an alternative assist gas instead of nitrogen.

Why compressed air?

- Contains approximately 78% nitrogen and 21% oxygen
- Oxygen provides exothermic support
- Nitrogen helps cooling and melt evacuation
- Can be generated on-site by compressor
- Reduces dependency on external nitrogen supply

Compressed Air = 78% N₂ + 21% O₂



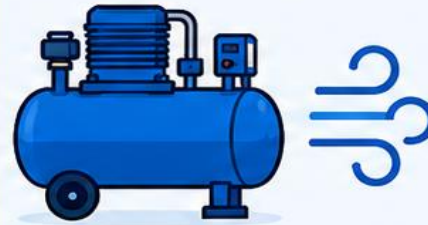
Cost Comparison: Nitrogen vs Compressed Air

Cost Item	Nitrogen Cutting	Compressed Air Cutting
Hourly Cost	570 TL/hour	98 TL/hour
Hourly Saving	—	472 TL/hour



Nitrogen
570 TL/h

$38 \text{ m}^3/\text{h} \times 15 \text{ TL}/\text{m}^3$



Compressed Air
98 TL/h

22 kW compressor, 90%
load factor, incl. maintenance



Saving: 472 TL/h



Annual Saving Calculation:

$472 \text{ TL}/\text{hour} \times 16 \text{ hours}/\text{day} \times 250 \text{ days}/\text{year}$

= 1,888,000 TL/year



Nitrogen cost was calculated from $38 \text{ m}^3/\text{h}$ consumption and $15 \text{ TL}/\text{m}^3$ unit cost. Compressed air cost was calculated based on a 22 kW compressor, 90% load factor, and maintenance allowance.

Solution 2: Parameter Optimization

Parameter		Optimized Approach
 1. Focus Position		Negative focus
 2. Gas Pressure		14–16 Bar
 3. Nozzle Type		Single nozzle
 4. Cutting Speed		Increased feed rate
 5. Assist Gas		Compressed air

The parameter optimization focuses on negative focus and high gas pressure. Negative focus helps the air enter the cutting kerf more effectively, while high pressure removes molten metal before it solidifies as dross.

ROI and Economic Impact

Investment Cost: 600,000 TL

Monthly Saving: 166,144 TL

ROI Period: $600,000 / 166,144 = 3.6$ months

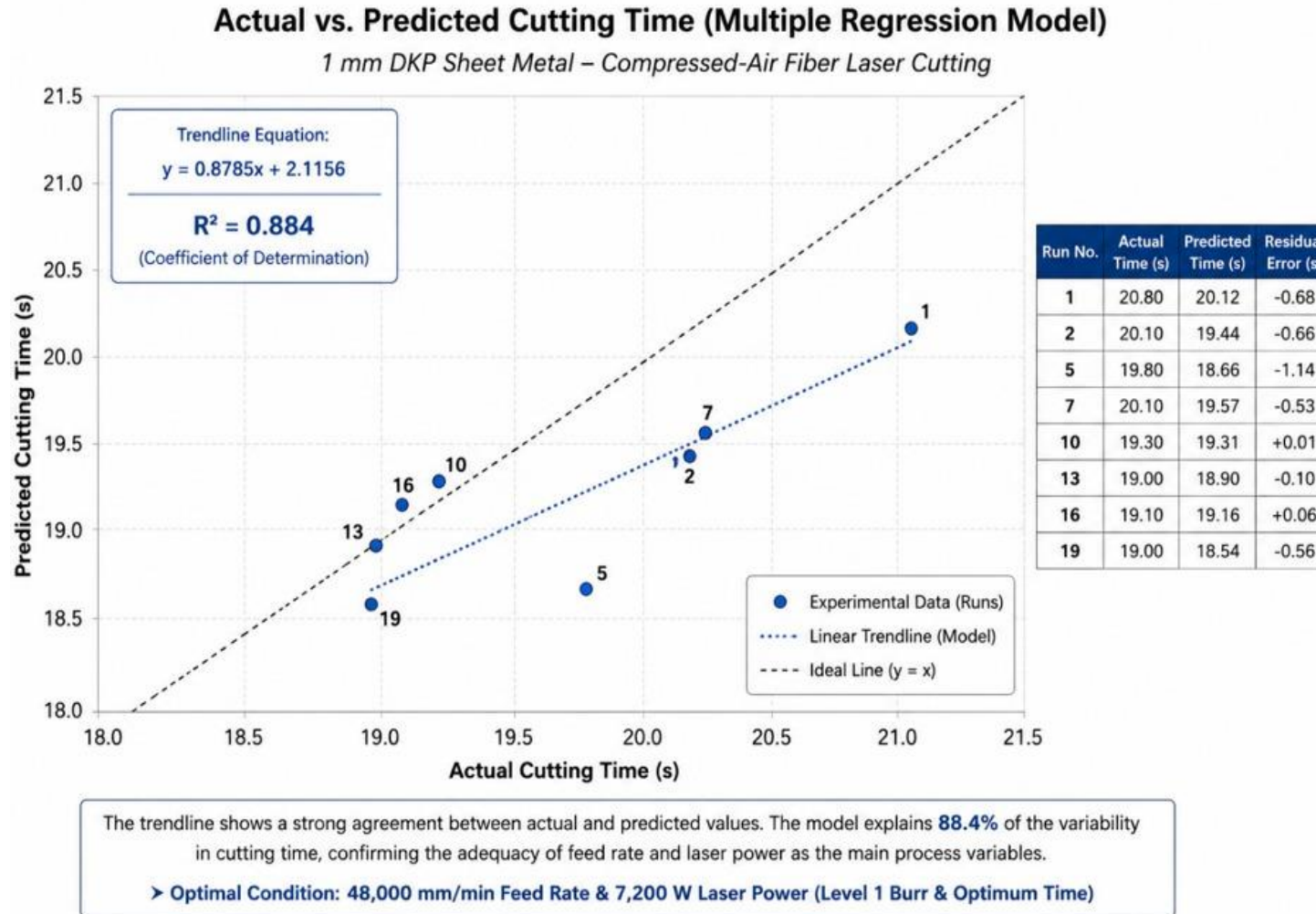
Economic Impact:

- Fast payback period
- Direct operating cost reduction
- Lower dependency on external gas suppliers
- High saving potential for multiple machines

Payback Timeline



DOE and Regression Analysis



Purpose of Regression Analysis:

- To compare actual and predicted cutting times
- To evaluate the effect of feed rate and laser power
- To verify the reliability of the optimized cutting parameters

Model Performance:

- $R^2 = 0.884$
- The model explains 88.4% of the variation in cutting time
- This indicates a strong agreement between actual and predicted values

The regression model confirms that feed rate and laser power are effective variables for predicting cutting time in 1 mm DKP sheet metal cutting.

Optimized Process Results

 Before	 After
 Nitrogen cutting	 Compressed air cutting
 20 m/min	 25 m/min
 Dross formation	 Low / no dross
 Manual deburring	 No heavy deburring
 High gas cost	 Lower operating cost

After the improvement, the process became faster and cleaner. The cutting speed increased by 25%, and the dross problem was minimized with optimized focus, pressure and nozzle selection.

Conclusion and Future Recommendations

This project demonstrated that compressed air cutting is technically feasible and economically beneficial for thin mild steel sheets.

Key Results

- 25% cycle time reduction
- Dross problem eliminated
- Manual deburring removed
- Assist gas cost reduced significantly
 - ROI achieved in 3.6 months

Future Recommendations

- Add optimized recipes to the machine process library
- Apply the method to other laser cutting machines
- Test higher thicknesses such as 4 mm and 6 mm
- Improve air filtration for long-term lens protection



**THANKS FOR
LISTENING**